
COMP0035 GROUP H REPORT

A PREPRINT

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1 Introduction

Our methodology in this project is inspired by the solution of March Machine Learning Mania 2025 [1], adapting proven strategies from large-scale sports prediction competitions to the context of football analytics. We aim to predict football match outcomes by combining statistical analysis and machine learning techniques. Our approach begins with comprehensive data preprocessing and feature engineering, including the calculation of team form, rolling averages, exponentially weighted moving averages, Elo ratings, head-to-head statistics and GLM-model-based team qualities. These features are designed to capture both short-term and long-term performance trends of teams.

For model training, we experiment with several algorithms: Random Forest, XGBoost, Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machines (SVM). Each model is evaluated using cross-validation and relevant metrics such as accuracy and ROC-AUC to ensure robust performance. Hyperparameter tuning is performed to optimise each algorithm. Among all models, the best accuracy achieved is approximately 55%, which represents a strong result given the inherent unpredictability and complexity of football match outcomes.

2 Headings: first level

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2.1 Headings: second level

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2.1.1 Headings: third level

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3 Examples of citations, figures, tables, references

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The documentation for natbib may be found at

<http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf>

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

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produces

Hasselmo, et al. (1995) investigated...

<https://www.ctan.org/pkg/booktabs>

3.1 Figures

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3.2 Tables

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3.3 Lists

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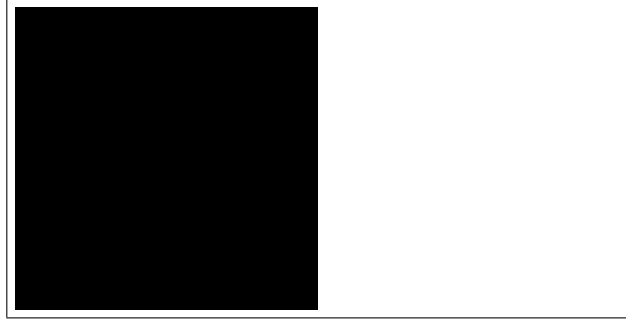


Figure 1: Sample figure caption.

Table 1: Sample table title

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Name	Description	Size (μm)
Dendrite	Input terminal	~ 100
Axon	Output terminal	~ 10
Soma	Cell body	up to 10^6

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4 Conclusion

This project explored statistical and machine learning methods for predicting football match outcomes, achieving a best accuracy of approximately 55%, which is notable given the inherent unpredictability of football results. Our results demonstrate that integrating domain knowledge with advanced modeling techniques can improve predictive performance.

Despite promising results, there is room for improvement. Future work could incorporate additional data sources, such as player statistics, injury reports, and advanced features including possession, tactical matchup analyses, and quantifiable psychological factors, as well as explore deep learning architectures and ensemble methods to further enhance predictive accuracy.

Overall, this project highlights the potential of data-driven approaches in football analytics and provides a foundation for future advancements in sports prediction.

References

- [1] Mohammad Odeh. March Machine Learning Mania 2025 Competition Solution. <https://www.kaggle.com/code/modeh7/final-solution-ncaa-2025/notebook>
- [2] George Kour and Raid Saabne. Fast classification of handwritten on-line arabic characters. In *Soft Computing and Pattern Recognition (SoCPaR), 2014 6th International Conference of*, pages 312–318. IEEE, 2014.
- [3] Guy Hadash, Einat Kermamy, Boaz Carmeli, Ofer Lavi, George Kour, and Alon Jacovi. Estimate and replace: A novel approach to integrating deep neural networks with existing applications. *arXiv preprint arXiv:1804.09028*, 2018.